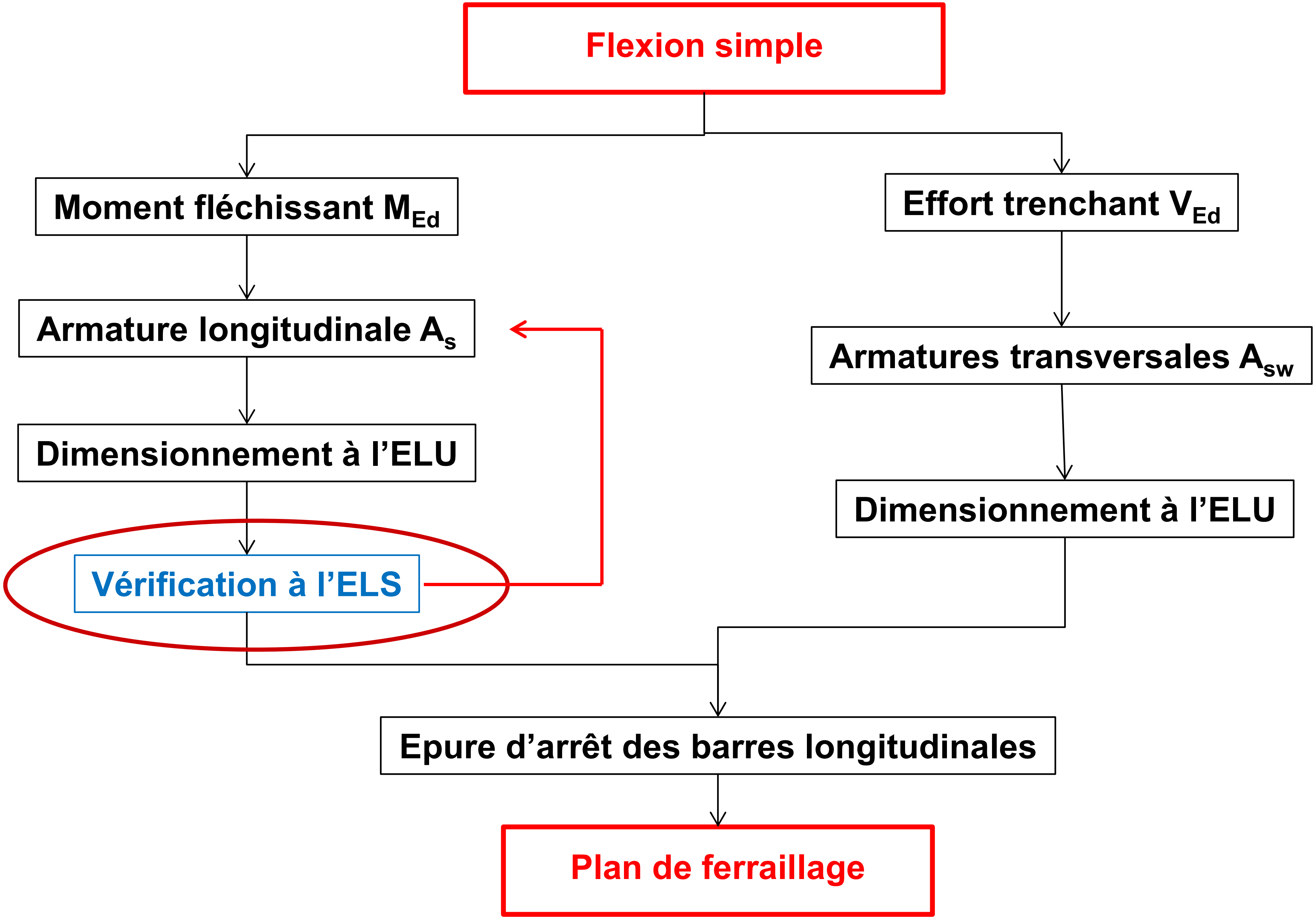
CHAPTER 6

Flexion simple à l’Etat limite de Service (ELS)

CONTENT

1 – Introduction



1 – Introduction

Requirement at Serviceability Limit State

- **Limitation des contraintes dans le béton σ_c et les armatures σ_s**
- **Maîtrise de la fissuration→ Limitation de l'ouverture des fissures**
- **Limitation des flèches**

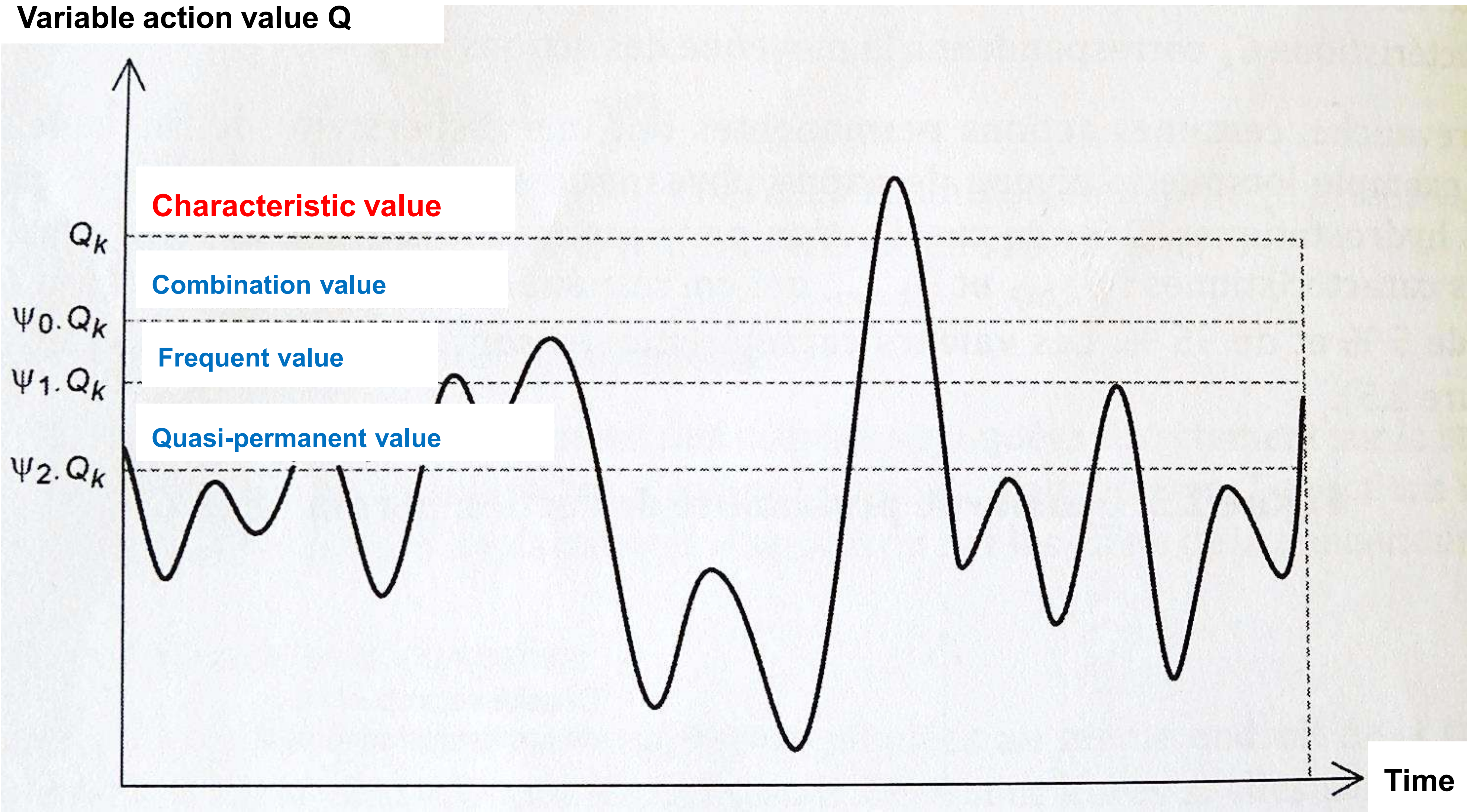
Elastic Bending of an under reinforced concrete beam

2 – Assumptions

- **Navier–Bernouilli theory: plane and straight sections remain plane before and after deformations → Linear strain diagram**
- **Perfect bond between reinforcement and concrete → Same strain of reinforcement and concrete at the same level**
- **Cracking → Concrete tensile strength neglected**
- **Linear behaviour of the materials →** $\sigma = E\,\varepsilon$
- **2 load combinations used:**
  - **Characteristic (M_{EC})** → $\sum_{j\geq 1} G_{k,j}$ «+» $Q_{k,1}$ «+» $\sum_{i>1} \psi_{0,i}\, Q_{k,i}$
  - **Quasi-permanent (M_{Eqp})** → $\sum_{j\geq 1} G_{k,j}$ «+» $\sum_{i\geq 1} \psi_{2,i}\, Q_{k,i}$

2 – Assumptions

Classify as a function of the characteristic value Q_k



D. Ricotier (2012). Dimensionnement des structures en béton selon l'Eurocode 2. Edition Le Moniteur, Paris, France.

3 – Stress limitation

Limiting values

$\sigma_c \rightarrow$ Eviter les niveaux élevés de compression

Loi de fluage linéaire

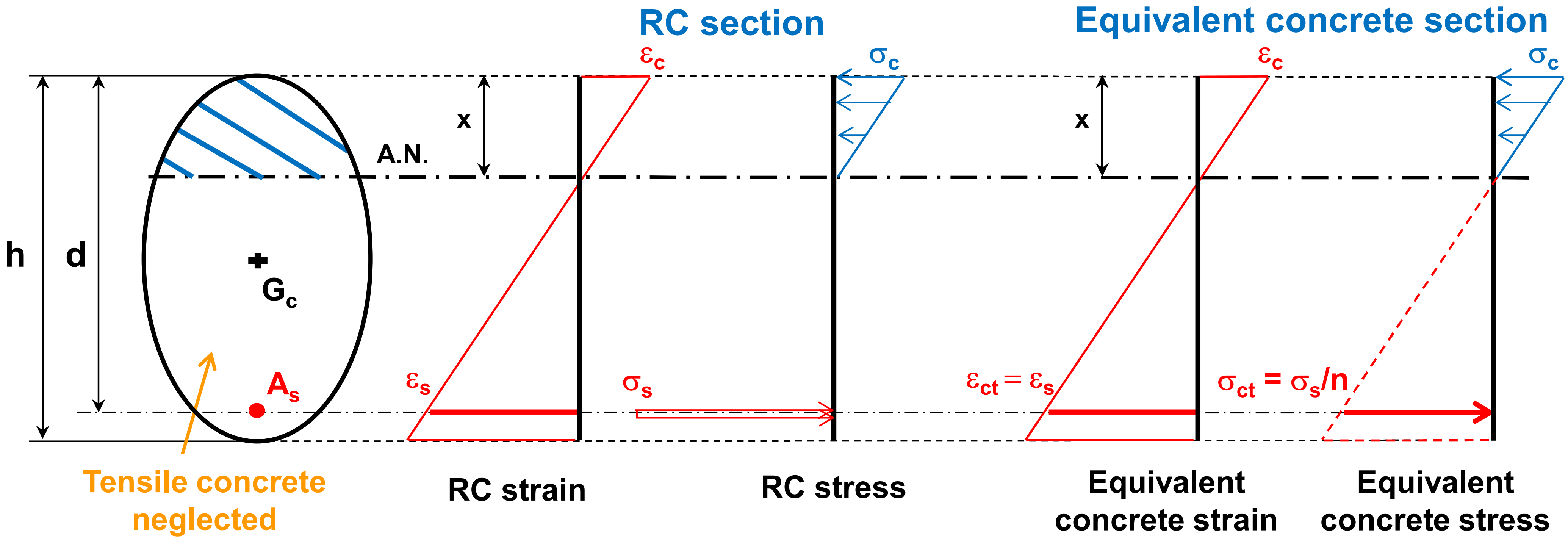
$\sigma_s \rightarrow$ Avoid inelastic strain, unacceptable cracking or deformation

	Concrete compressive stress	Tension stress in reinforcement
SLS characteristic	$\sigma_c \leq 0.6\ f_{ck}$ (Only for exposure classes XD, XF, XS)	$\sigma_s \leq 0.8\ (f_{yk}$ or $f_{pk})$
SLS quasi-permanent	$\sigma_c \leq 0.45\ f_{ck}$	-

Si non vérifié, fluage non-linéaire

3 – Stress limitation

Steel / concrete coefficient of equivalence n



$$\varepsilon_{ct} = \varepsilon_s \text{ where } \sigma_{ct} = \varepsilon_{ct} E_{c,eff}$$

$$\sigma_s = \varepsilon_s E_s$$

$$\Rightarrow \sigma_{ct} = \frac{\sigma_s}{\frac{E_s}{E_{c,eff}}} \Rightarrow \boxed{n = \frac{E_s}{E_{c,eff}}}$$

Usually: $15 \leq n \leq 20 \rightarrow$ Conservative value $n = 15$

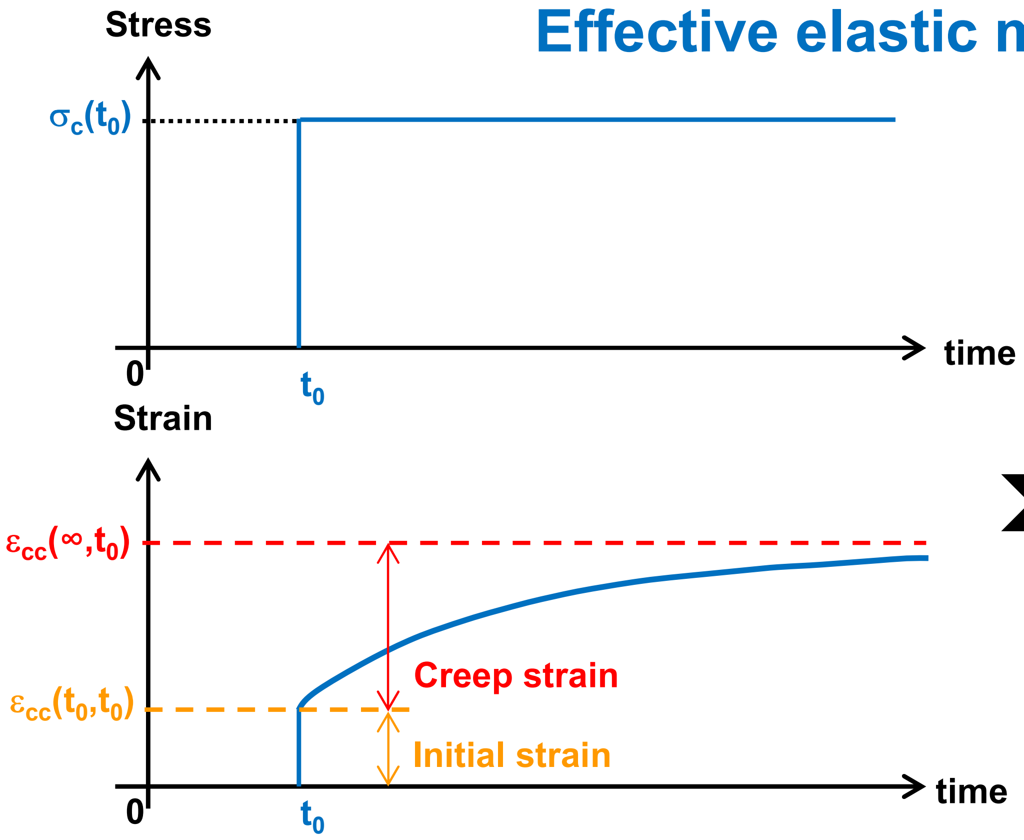
ε_{ct} : equivalent tensile strain in concrete / ε_s : tensile strain in reinforcement

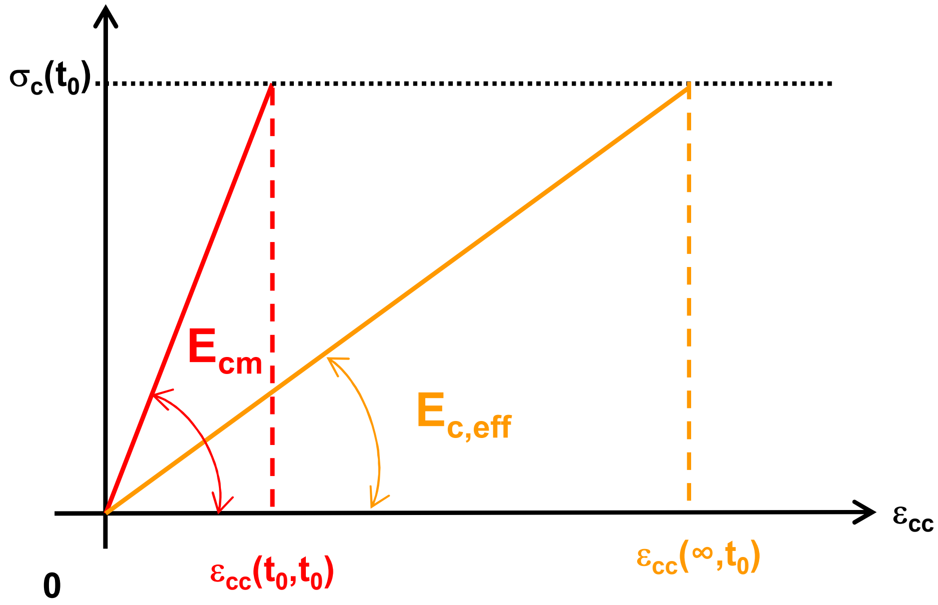
σ_{ct} : equivalent tensile stress in concrete/ σ_s : tensile stress in reinforcement

E_s : elastic modulus of reinforcing steel / $E_{c,eff}$: effective elastic modulus of concrete

3 – Stress limitation

Effective elastic modulus of concrete $E_{c,eff}$





$\sigma_c \leq 0.45\ f_{ck}(t_0)$

Linear creep function

$$\varepsilon_{cc}(\infty, t_0) = \varphi(\infty, t_0)\,\frac{\sigma_c}{E_c}$$

Usually: $2 \leq \varphi(\infty,t_0) \leq 3$

$\sigma_c \leq 0.45\ f_{ck}(t_0)$

Linear creep function

$$E_{c,eff} = \frac{E_{cm}}{1 + \varphi_{ef}}$$

$$\varphi_{ef} = \varphi(\infty,t_0)\,\frac{M_{Eqp}}{M_{EC}}$$

3 – Stress limitation

Effective elastic modulus of concrete $E_{c,eff}$

$$\varphi_{ef} = \varphi(\infty, t_0)\frac{M_{Eqp}}{M_{Ed,ELS}} \tag{8.3}$$

M_{Eqp} : est le moment dû aux charges quasi-permanentes, $M_{Eqp} = G + \psi_2 Q$ (habituellement $\psi_2 = 0,3$, voir tableau 2.3)

$M_{Ed,ELS}$: est le moment sollicitant. Pour les justifications classiques, la combinaison est celle caractéristique des charges donc $M_{Ed,ELS} = G + Q$ (pour une seule charge permanente et une seule charge d'exploitation).

Remarque

Le terme φ_{ef} permet donc de moduler l'impact du chargement en fonction de leur durée sur le fluage. Il réduit le coefficient de fluage finale pour permettre de prendre en compte les charges des actions **à court terme** (charge d'exploitation) et les charges des actions **à long terme** (charge du poids propre).

φ_{ef} est à utiliser avec la combinaison caractéristique des actions.

3 – Stress limitation

Final creep coefficient $\varphi(\infty, t_0)$

70 years → Accuracy = ± 20 %

Cement of class R:

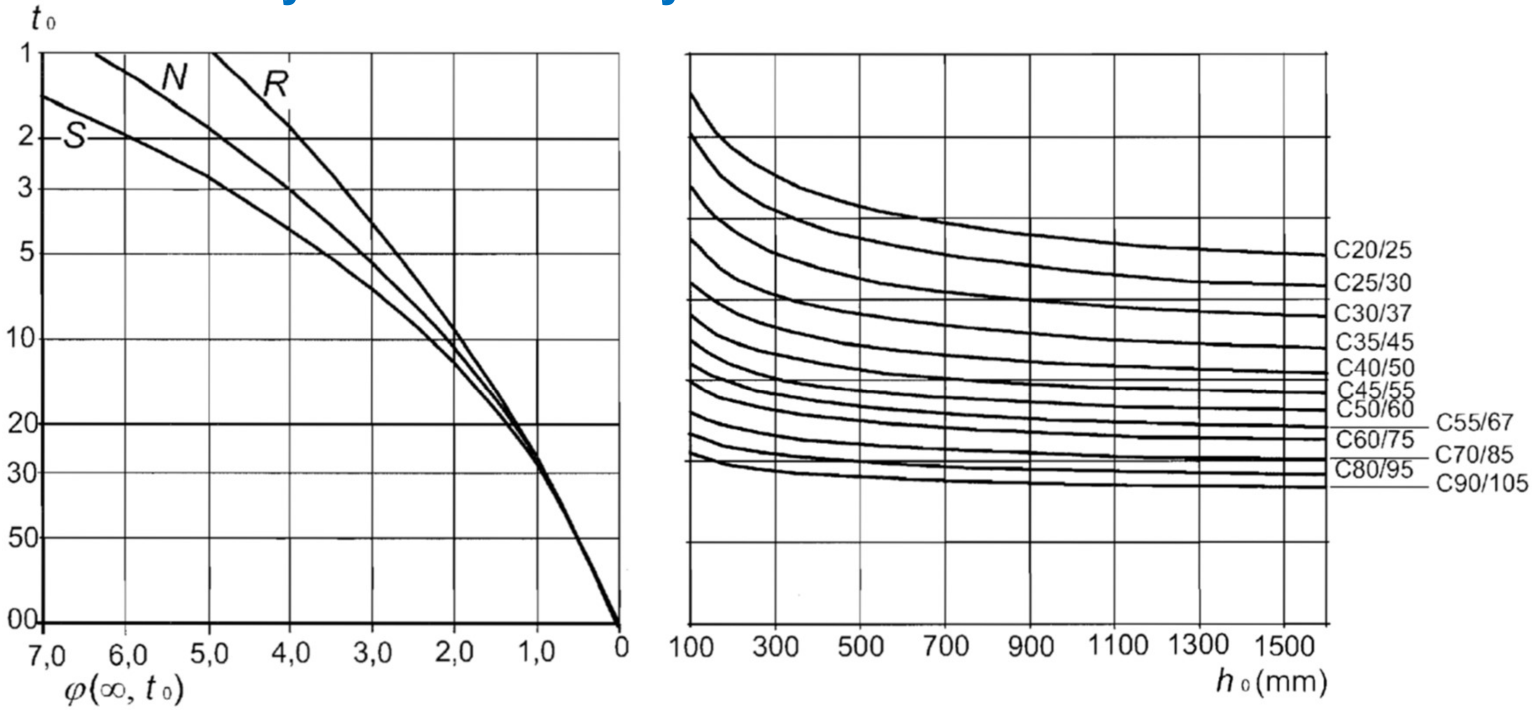
- **CEM 42.5R**
- **CEM 52.5N**
- **CEM 52.5R**

Cement of class N:

- **CEM 32.5R**
- **CEM 42.5N**

Cement of class S:

- **CEM 32.5N**



a) inside conditions - RH = 50%

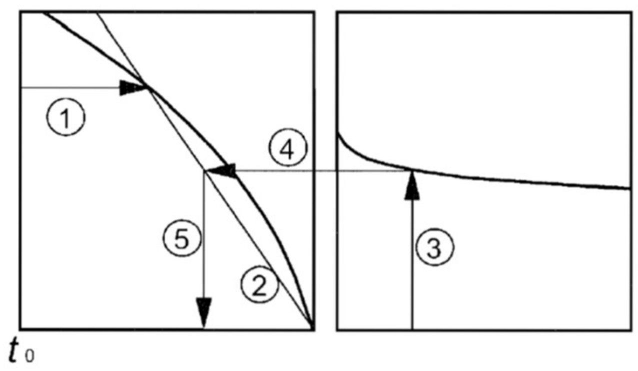
t_0 = age of concrete at time of loading (j)

h_0 = notional size

$$h_0 = \frac{2.A_c}{u}$$

A_c = concrete cross-sectional area

u = perimeter exposed to drying



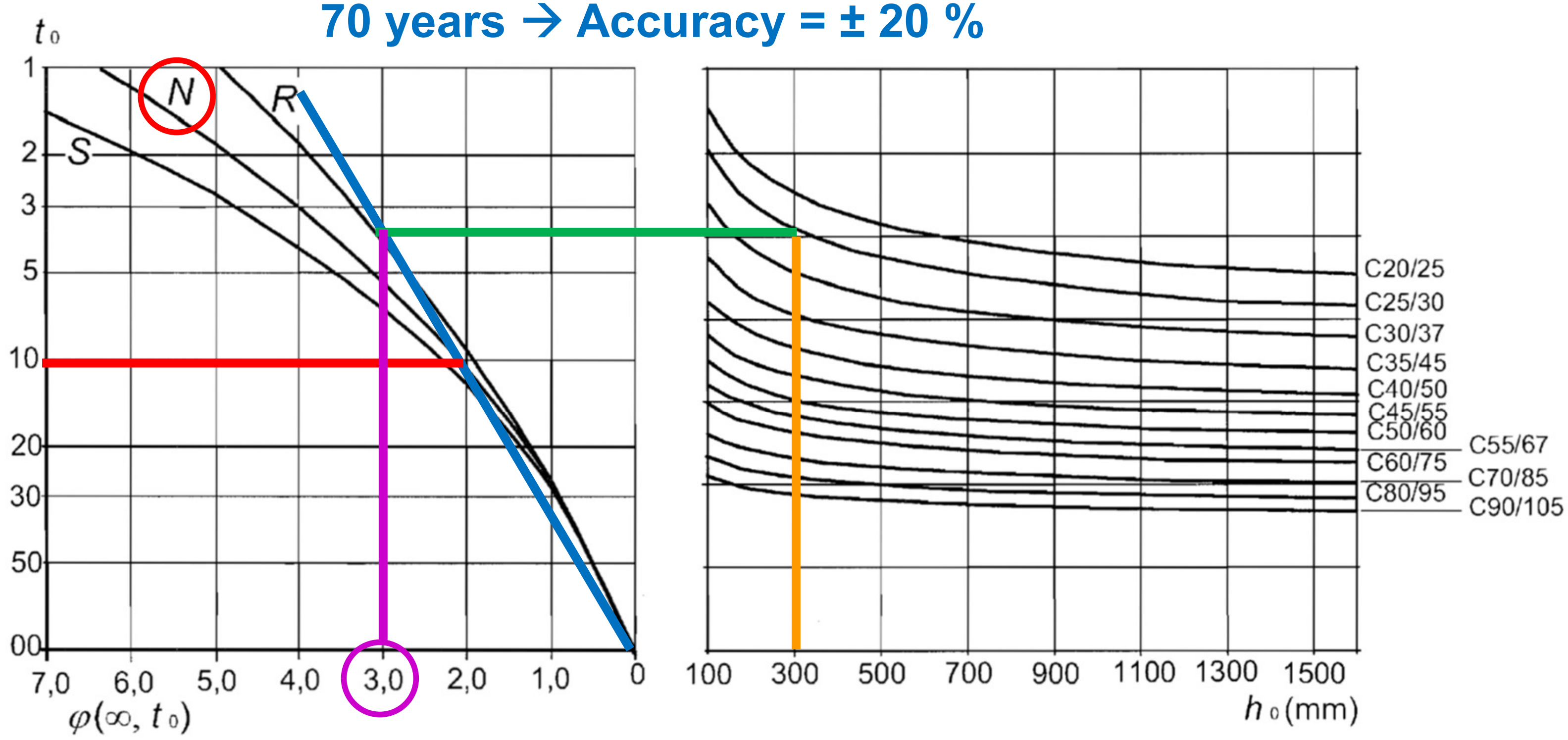
Note:
- intersection point between lines 4 and 5 can also be above point 1
- for t_0 > 100 it is sufficiently accurate to assume t_0 = 100 (and use the tangent line)

NF EN 1992 (December 2004). Eurocode 2 – Design of concrete structures. CEN

3 – Stress limitation

Final creep coefficient $\varphi(\infty,t_0)$

70 years → Accuracy = ± 20 %

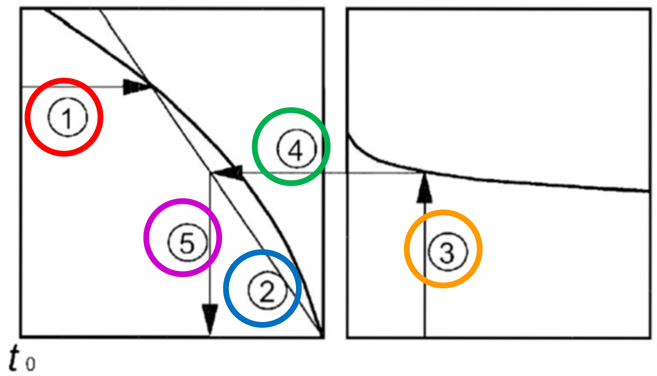


a) inside conditions - RH = 50%

Example:

t_0 = 10 j
h_0 = 300 mm
C25/30
Class N

$\boxed{\varphi(\infty,t_0) = 3}$



Note:
- intersection point between lines 4 and 5 can also be above point 1
- for t_0 > 100 it is sufficiently accurate to assume t_0 = 100 (and use the tangent line)

NF EN 1992 (December 2004). Eurocode 2 – Design of concrete structures. CEN

3 – Stress limitation

$\varphi(\infty,t_0)$ est le coefficient de fluage final $\varphi(\infty,t_0) = \varphi_0$

φ_0 est le coefficient de fluage conventionnel et peut-être estimé par :

$$\varphi_0 = \varphi_{RH} \times \beta(f_{cm}) \times \beta(t_0)$$

φ_{RH} est un facteur tenant compte de l'influence de l'humidité relative sur le coefficient de fluage conventionnel,

$$\varphi_{RH} = \begin{cases} 1 + \dfrac{1 - RH/100}{0{,}1\,\sqrt[3]{h_0}} & f_{cm} \leq 35 \text{ MPa} \\ \left[1 + \dfrac{1 - RH/100}{0{,}1\,\sqrt[3]{h_0}} a_1\right] a_2 & f_{cm} > 35 \text{ MPa} \end{cases}$$

avec $a_1 = \left(\dfrac{35}{f_{cm}}\right)^{0,7}$, $a_2 = \left(\dfrac{35}{f_{cm}}\right)^{0,2}$ et $h_0 = \dfrac{b_w.h}{b_w + h}$ (h_0 est le rayon moyen de l'élément en **mm**, c'est-à-dire le rapport entre $2\dfrac{A_c}{P}$, avec A_c l'aire de la section droite et P le périmètre de l'élément en contact avec l'atmosphère).

f_{cm} est en **MPa**.

$$\beta(f_{cm}) = \frac{16{,}8}{\sqrt{f_{cm}}}$$

$$\beta(t_0) = \frac{1}{0{,}1 + t_0^{0,2}}$$

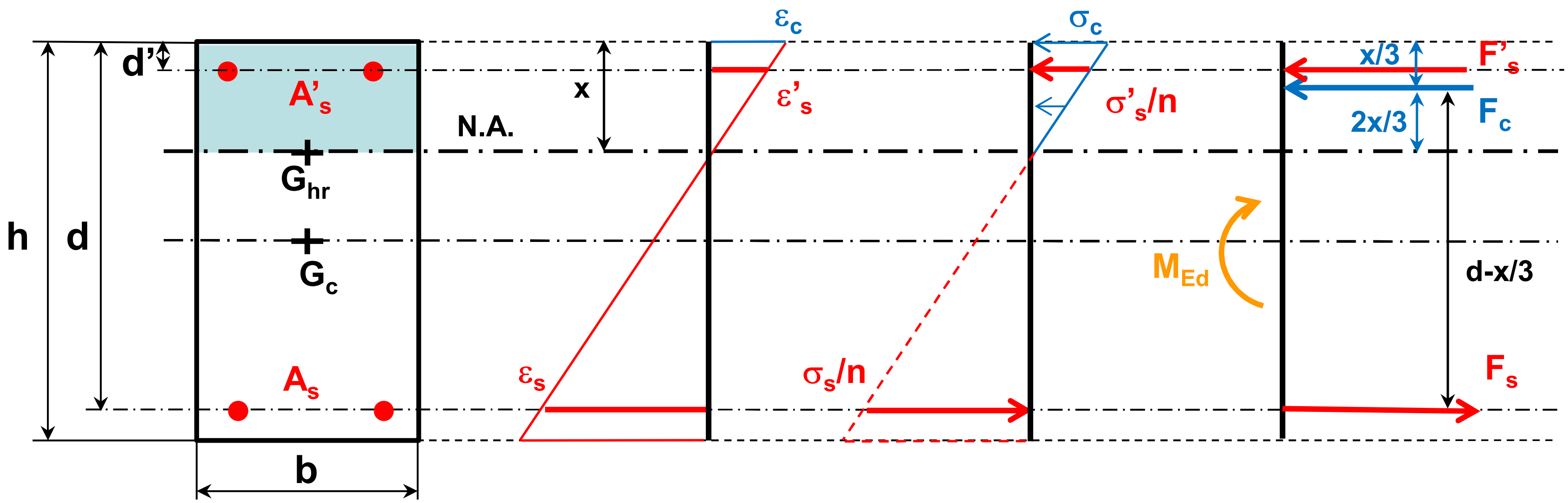
$\beta(f_{cm})$ est un facteur tenant compte de l'influence de la résistance du béton sur le coefficient de fluage conventionnel,

3 – Stress limitation

$\beta(t_0)$ est un facteur tenant compte de l'influence de l'âge du béton au moment du chargement sur le coefficient de fluage conventionnel,
t_0 est l'âge du béton au moment du chargement, en jour (habituellement t_0 = 28 jours).

3 – Stress limitation

Rectangular section



Equivalent concrete cross sectional area section

$$A_s \rightarrow n\,A_s$$

$$A'_s \rightarrow n\,A'_s$$

3 – Stress limitation

Rectangular section

Inputs

M_{Ed}, b, h, d, d', n, A_s, A'_s

↓

1^{st} moment of area from N.A.

$$\frac{bx^2}{2} + n\,A'_s\,(x - d') - n\,A_s\,(d - x) = 0 \rightarrow x$$

↓

2^{nd} moment of area from N.A.

$$I_{hr} = \frac{bx^3}{3} + n\,A'_s\,(x - d')^2 + n\,A_s\,(d - x)^2$$

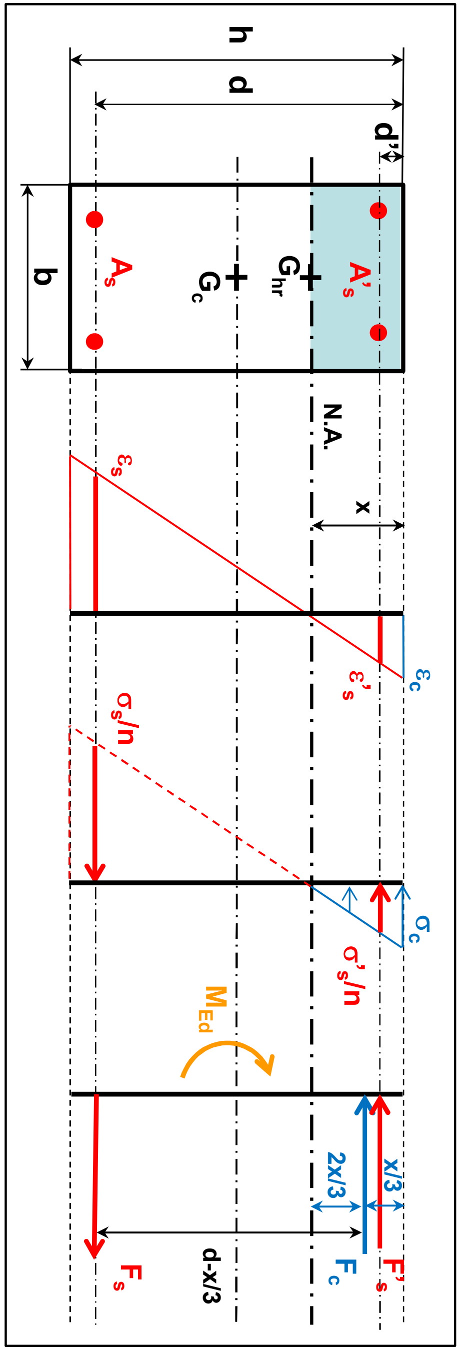
↓

Tensile stress in reinforcement

$$\sigma_s = \frac{n\,M_{Ed}\,(d - x)}{I_{hr}}$$

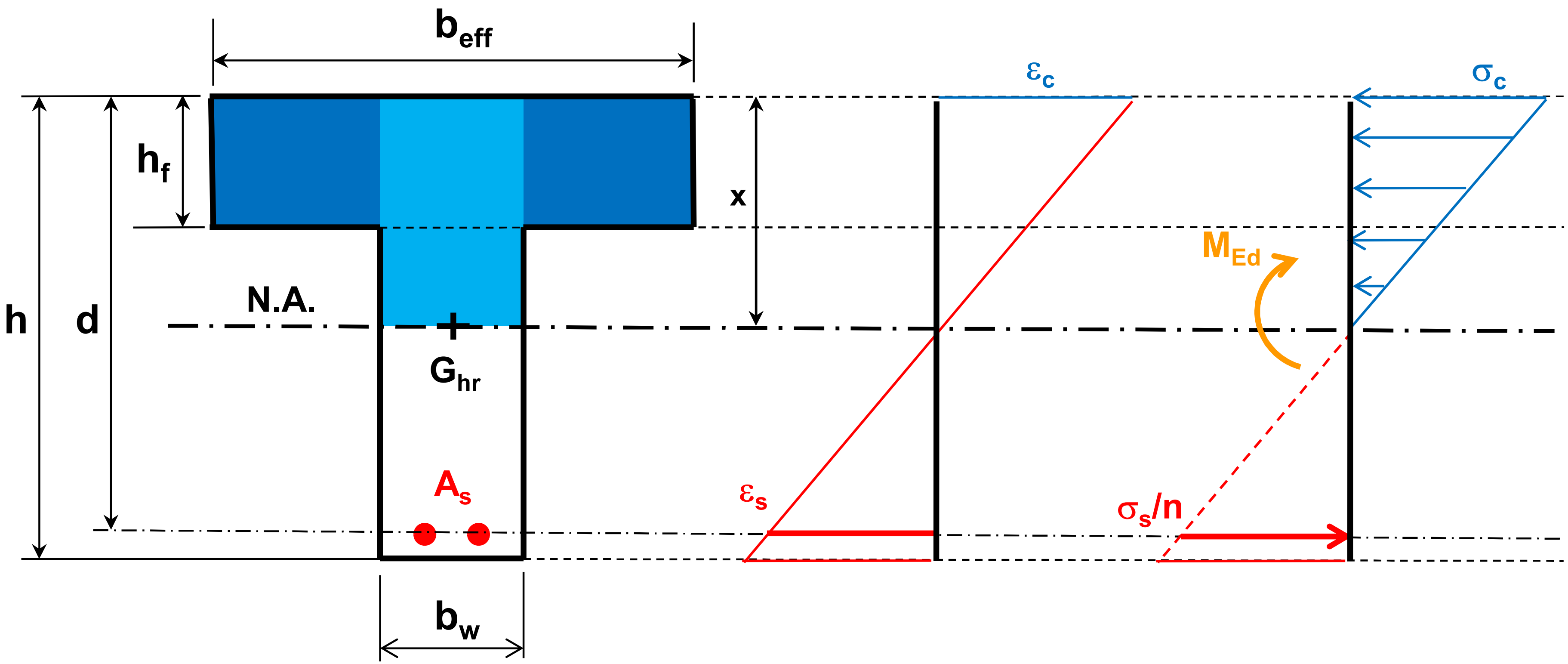
Compressive stress in concrete

$$\sigma_c = \frac{M_{Ed}\,x}{I_{hr}}$$



3 – Stress limitation

T-section without A'_s



Equivalent concrete cross sectional area section

$$A_s \rightarrow n\, A_s$$

3 – Stress limitation

T-section without A'_s

Inputs

$M_{Ed}, b_{eff}, b_w, h, h_f, d, A_s$

↓

1st moment of area from N.A.

$$\frac{b_w\, x^2}{2} + (b_{eff} - b_w)\, h_f \left(x - \frac{h_f}{2}\right) - n\, A_s\, (d - x) = 0 \rightarrow x$$

↓

$$x \geq h_f$$

NO → **Design as a rectangular section (b_{eff} h)**

YES ↓

2nd moment of area from N.A.

$$I_{hr} = \frac{b_w\, x^3}{3} + (b_{eff} - b_w)\frac{h_f^3}{12} + (b_{eff} - b_w)\, h_f \left(x - \frac{h_f}{2}\right)^2 + n\, A_s\, (d - x)^2$$

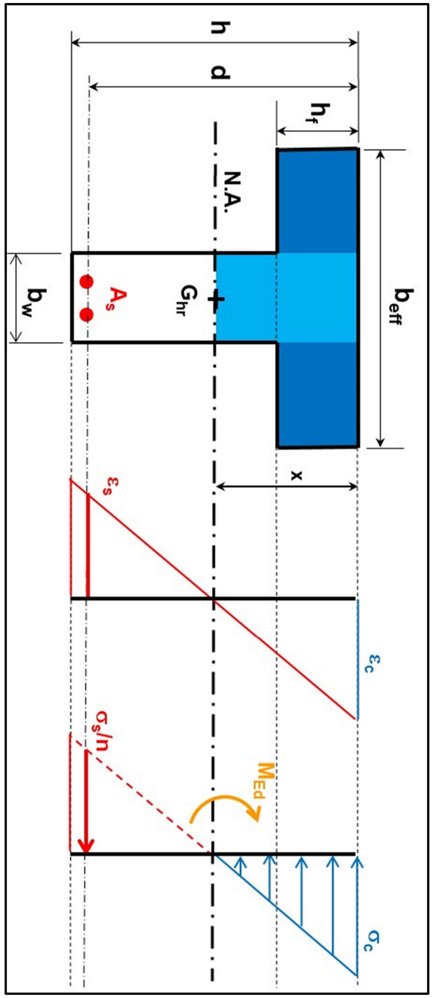
↓

Tensile stress in reinforcement

$$\sigma_s = \frac{n\, M_{Ed}\, (d - x)}{I_{hr}}$$

Compressive stress in concrete

$$\sigma_c = \frac{M_{Ed}\, x}{I_{hr}}$$



3 – Stress limitation

T-section with A'_s

Inputs

$M_{Ed}, b_{eff}, b_w, h, h_f, d, d', A_s, A'_s$

↓

1st moment of area from N.A.

$$\frac{b_w\, x^2}{2} + (b_{eff} - b_w)\, h_f \left(x - \frac{h_f}{2}\right) - n\, A_s\, (d - x) + n\, A'_s\, (x - d') = 0 \rightarrow x$$

↓

$x \geq h_f$

NO → **Design as a rectangular section (b_{eff} h)**

YES ↓

2nd moment of area from N.A.

$$I_{hr} = \frac{b_w\, x^3}{3} + (b_{eff} - b_w)\frac{h_f{}^3}{12} + (b_{eff} - b_w)\, h_f \left(x - \frac{h_f}{2}\right)^2 + n\, A_s (d - x)^2 + n\, A'_s (x - d')^2$$

Tensile stress in reinforcement

$$\sigma_s = \frac{n\, M_{Ed}\, (d - x)}{I_{hr}}$$

Compressive stress in concrete

$$\sigma_c = \frac{M_{Ed}\, x}{I_{hr}}$$

4 – Crack control

Minimum reinforcement area

Parameters taking into account stress distribution and loading

Maximum effort in tensile zone

$$\boxed{A_{s,min} = \frac{k_c\, k\, f_{ct,eff}\, A_{ct}}{f_{yk}}}$$

A_{ct} = aire de béton tendu juste avant la formation de la première fissure (A_s négligé)

Rectangular cross section → A_{ct} = b h/2

$f_{ct,eff} = f_{ct,m}$

k → repartition non linéaire des contraintes de traction f_{ct} sur h (section rectangulaire ou section en T avec l'âme tendu) ou b_{eff} (section en T avec la membrane tendue)

k = 1 for (h or b_{eff}) ≤ 300 mm

k = 0.65 for (h or b_{eff}) ≥ 800 mm

k = 1.21 – 0.7×(h or b_{eff}) for 300 < h < 800 mm

k_c → section non fissure sur toute la zone tendue

k_c = 0.4 pour flexion simple

4 – Crack control

Minimum reinforcement area

Type of loading	Rectangular sections and web of T-sections and box sections	Flanges of T-sections and box sections
Pure tensile	$k_c = 1$	$k_c = 1$
Bending or combined bending and axial load	$k_c = 0{,}4.\left[1 - \dfrac{\sigma_c}{k_1.\left(\frac{h}{h^*}\right).f_{ct,eff}}\right] \leq 1$	$k_c = 0{,}9.\dfrac{F_{cr}}{A_{ct}.f_{ct,eff}} \geq 0{,}5$

where

σ_c : Mean compressive stress in concrete : $\sigma_c = \frac{N_{Ed}}{b.h}$, where N_{Ed} is the axial force at SLS (positive in compression);

$k_1 =$ 1.5 if N_{Ed} is a compressive force and $k_1 = \frac{2}{3}.\frac{h^*}{h}$ if N_{Ed} is a tensile force ;

$h^* = Min\{h, 1\text{ m}\}$;

F_{cr} : Absolute value of the tensile force within the flange immediately prior cracking due to the crack moment calculated with $f_{ct,eff}$

D. Ricotier (2012). Dimensionnement des structures en béton selon l'Eurocode 2. Edition Le Moniteur, Paris, France.

4 – Crack control

Ouverture de fissure w_k

Bon fonctionnement de la structure

Aspect innaceptable

Quasi-permanent combination of loads → $w_k \leq w_{max}$

w_{max} = limiting values of the crack width

Reinforced Concrete

w_{max} limiting values

(in mm)

Valeurs recommandées de w_{max} suivant l'AN française, *(ANF 7.1)*.

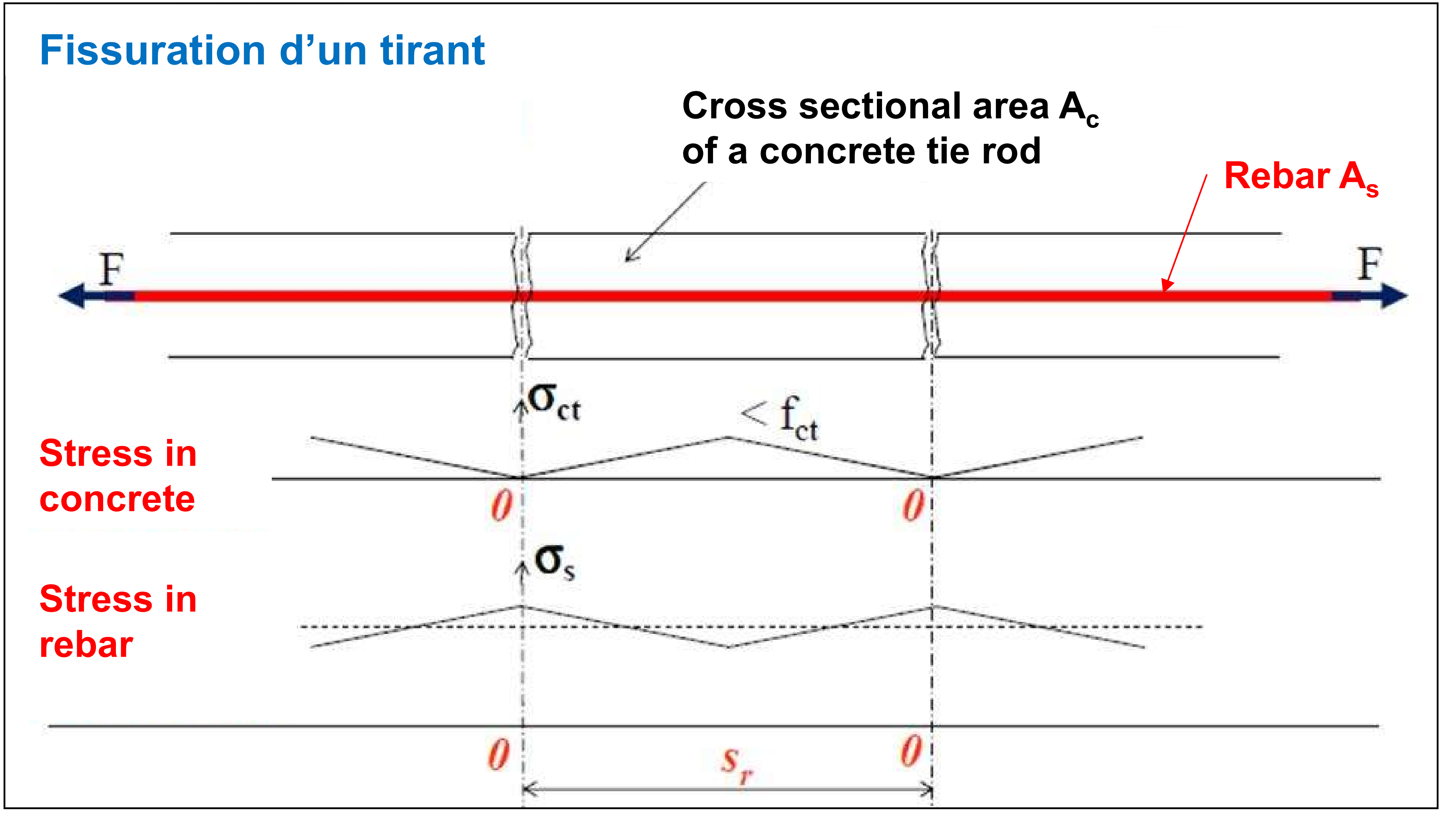
Classe d'exposition	Éléments en béton armé et éléments en béton précontraint sans armatures adhérentes	Éléments en béton précontraint avec armatures adhérentes
	Combinaison quasi-permanente de charges	Combinaison fréquente de charges
X0, XC1	0,40[a]	0,20[a]
XC2, XC3, XC4	0,30[b]	0,20[c]
XD1, XD2, XS1, XS2, XS3, XD3	0,20	Décompression

[a] sauf demande spécifique des documents particuliers du marché, le calcul de w_{max} n'est pas requis si les dispositions constructives autres que celles du présent chapitre sont respectées ;
[b] comme précédemment pour les bâtiments des catégories d'usage A à D (voir tableau n°2.3)
[c] il convient en plus de vérifier la décompression

4 – Crack control

Crack width w_k

Fissuration d'un tirant

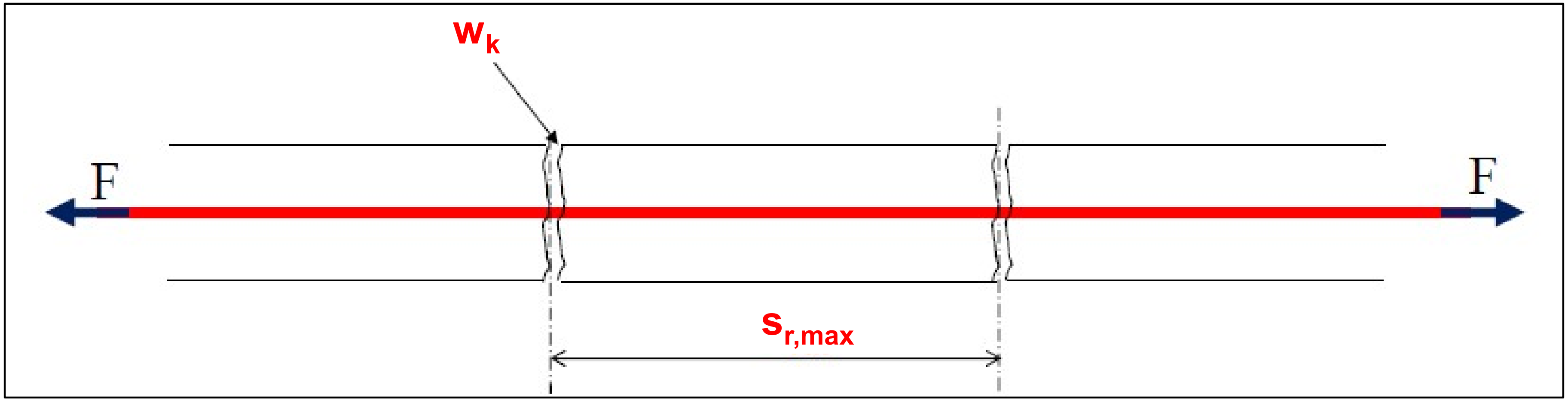


S. Multon (2012). Cours de Béton armé – Eurocode 2 – INSA Toulouse.

$$F = A_c\,\sigma_{ct} + A_s\,\sigma_s$$

4 – Crack control

Crack width w_k



Crack width

$$w_k = s_{r,max} \, (\varepsilon_{sm} - \varepsilon_{cm})$$

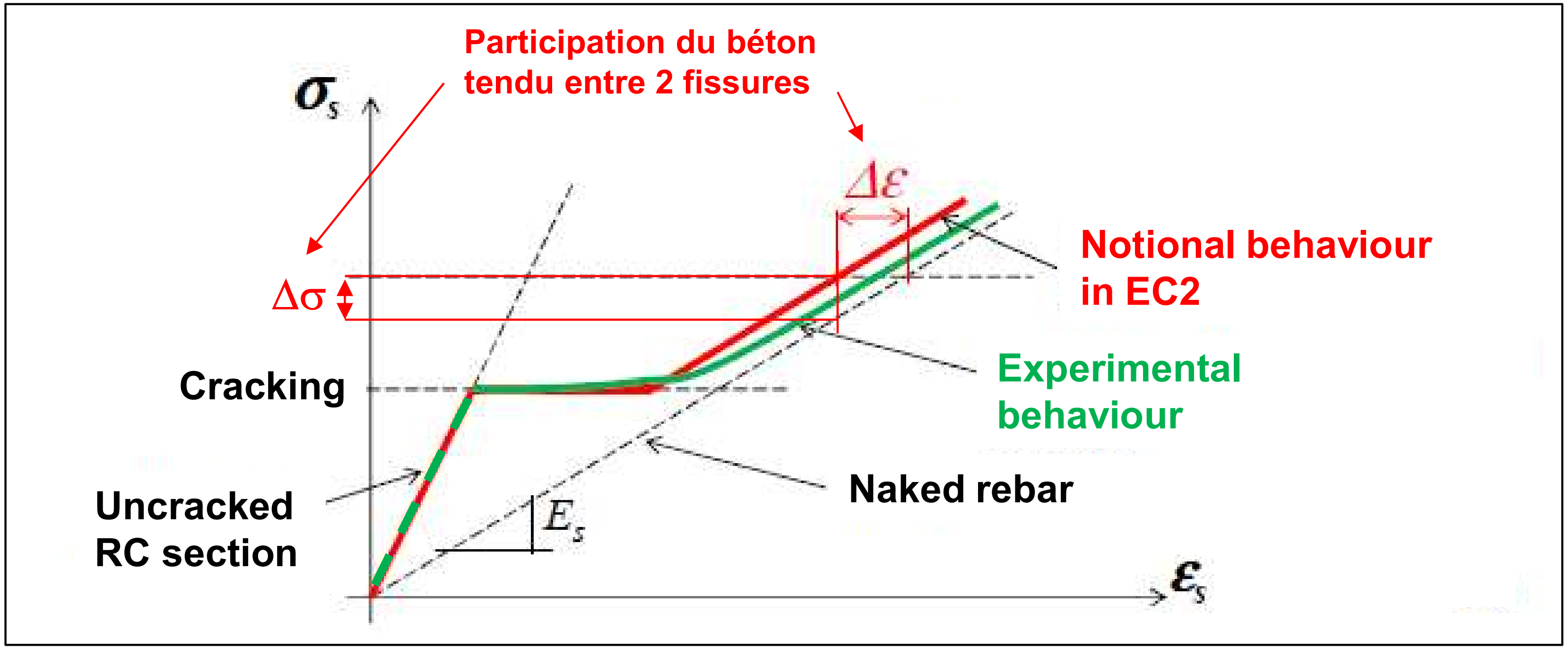
Where $s_{r,max}$ = maximum crack spacing

ε_{sm} = mean strain in reinforcement

ε_{cm} = mean strain in concrete between cracks

4 – Crack control

Crack width w_k - Mean strains ε_{sm} and ε_{sm}



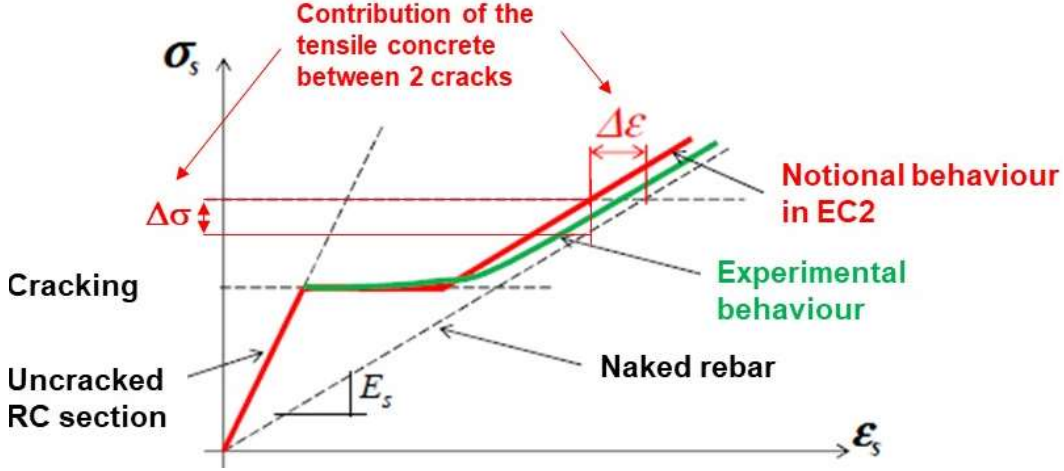
S. Multon (2012). Cours de Béton armé – Eurocode 2 – INSA Toulouse.

$$\Delta\varepsilon = \textbf{strain variation in reinforcement}$$

$$\Delta\sigma = \textbf{stress variation in reinforcement}$$

4 – Crack control

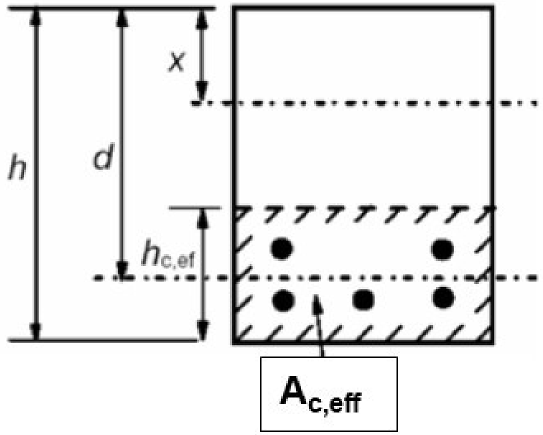
Crack width w_k - Mean strains ε_{sm} and ε_{sm}



Transmitted load by bond

$$\Delta F_s = k_t A_{c,eff} \, f_{ct,eff}$$

Load time



Strain variation in the rebar

$$\Delta\varepsilon = \frac{\Delta\sigma}{E_s} = \frac{\Delta F_s}{A_s E_s} = k_t \frac{f_{ct,eff}}{E_s \rho_{p,eff}} \quad \text{where} \quad \rho_{p,eff} = \frac{A_s}{A_{c,eff}}$$

Mean strain in reinforcement

$$\varepsilon_{sm} = \varepsilon_s - \Delta\varepsilon$$

$$\varepsilon_{sm} = \frac{\sigma_s}{E_s} - k_t \frac{f_{ct,eff}}{E_s \rho_{p,eff}}$$

Mean strain in concrete between 2 cracks

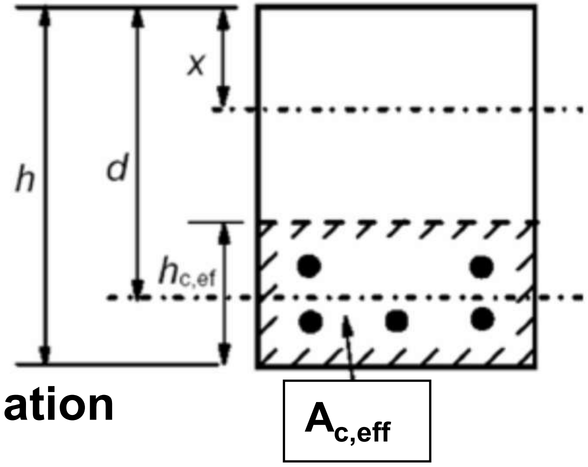
$$\varepsilon_{cm} = \frac{\Delta F_s}{A_{c,eff} E_{cm}} = \frac{k_t f_{ct,eff}}{E_{cm}}$$

$$\varepsilon_{cm} = \frac{k_t f_{ct,eff} \alpha_e}{E_s} \quad \text{where} \quad \alpha_e = \frac{E_s}{E_{cm}}$$

4 – Crack control

Crack width w_k - Mean strains ε_{sm} and ε_{sm}

$$\boxed{\varepsilon_{sm} - \varepsilon_{cm} = \frac{\sigma_s - k_t \frac{f_{ct,eff}}{\rho_{p,eff}}(1+\alpha_e\,\rho_{p,eff})}{E_s} \geq 0.6\,\frac{\sigma_s}{E_s}}$$



σ_s = stress in reinforcement under the quasi-permanent combination

$f_{ct,eff} = f_{ct,m}$

k_t = 0.6 (short term loading) - k_t = 0.4 (long term loading → common case)

$\alpha_e = E_s / E_{cm}$

$\rho_{p,eff} = A_s / A_{c,eff}$

$h_{c,ef}$ = Min {2.5 (h-d) ; (h-x)/3 ; h/2}

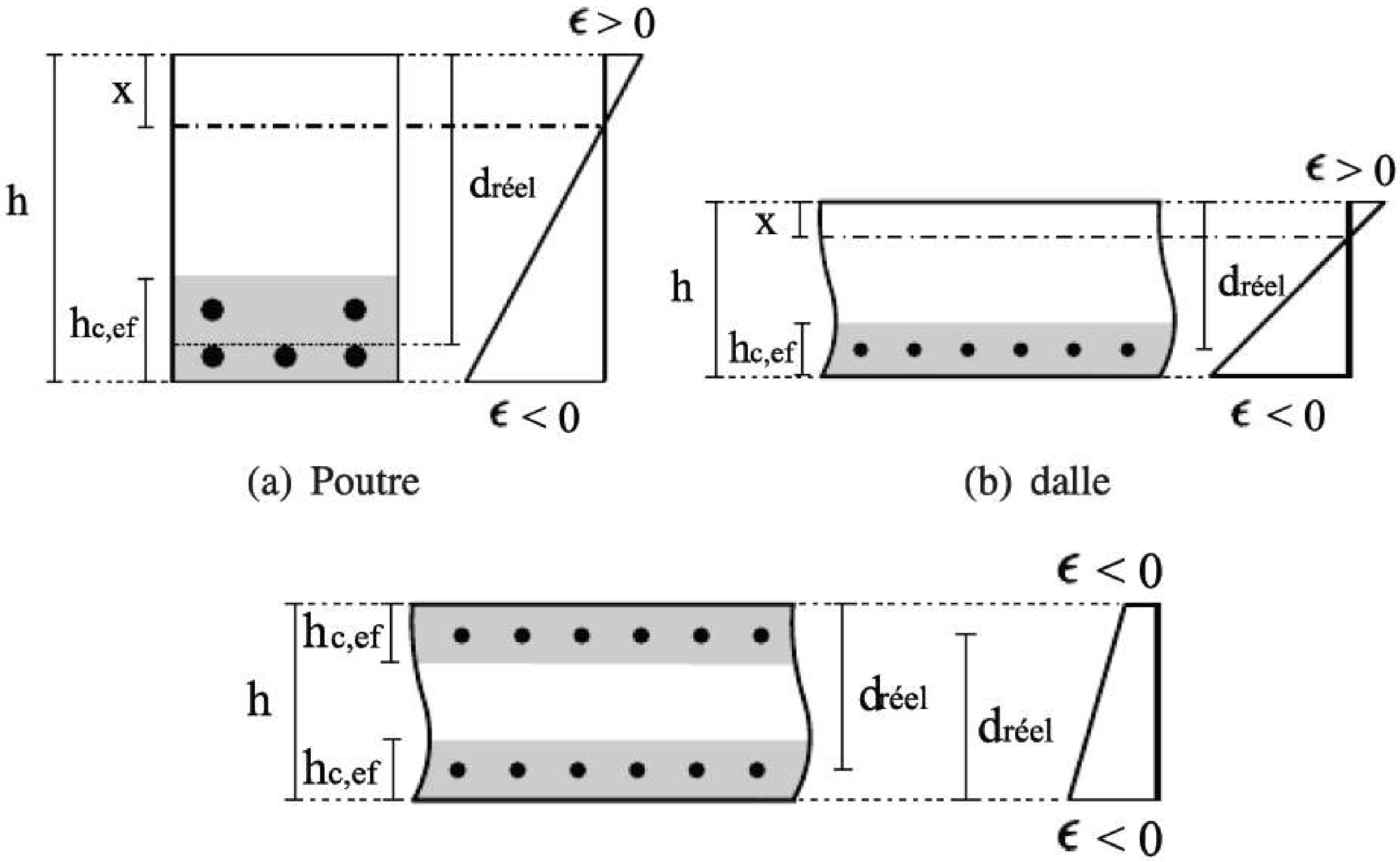
x = height of concrete in compression for a cracked homogeneous section

$A_{c,eff}$ → area of concrete surrounding the reinforcement

$A_{c,eff} = b\,h_{c,ef}$

4 – Crack control

Crack width w_k - Mean strains ε_{sm} and ε_{sm}



(a) Poutre (b) dalle

(c) Élément sollicité en traction

Sections effectives de béton autour des armatures tendues.

NF EN 1992 (December 2004). Eurocode 2 – Design of concrete structures. CEN .

4 – Crack control

Crack width w_k - Maximum crack spacing $s_{r,max}$

For a beam: spacing of reinforcement ≤ 5 [c+0.5 ϕ] → $s_{r,max} = k_3\, c + \dfrac{k_1\, k_2\, k_4\, \Phi_{eq}}{\rho_{p,eff}}$

Dégradation du béton

Adhérence acier/béton

c = enrobage des armatures

k_1 → propriétés d'adhérénce

k_1 = 0.8 (barres HA)

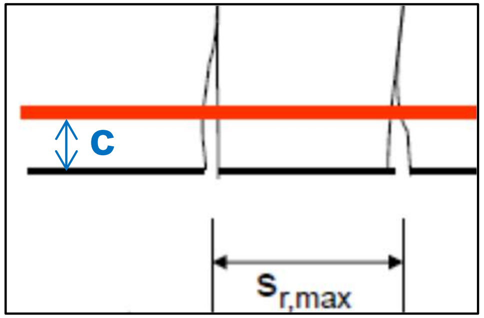
k_2 → distribution des déformations

k_2 = 0.5 (flexion simple ou composée)

k_3 = 3.4 pour c ≤ 25 mm

If not $k_3 = 3.4\,(25/c)^{2/3}$ where c in mm

k_4 = 0.425



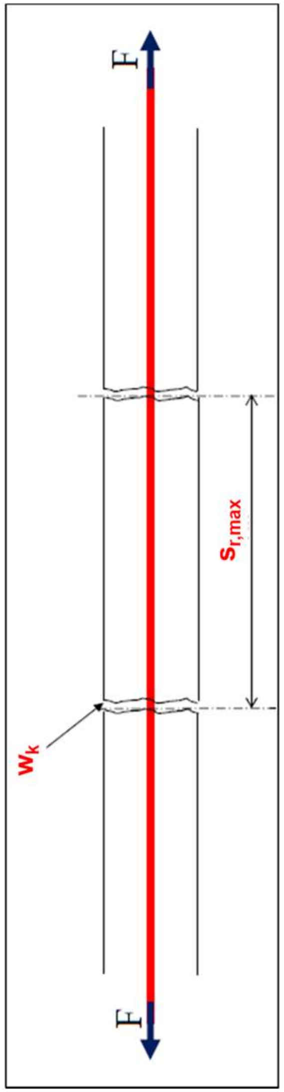
ϕ_{eq} = diameter equivalent des barres

For a section with n_i bars of diameter ϕ_i

$$\Phi_{eq} = \frac{\sum n_i\, \Phi_i^{\,2}}{\sum n_i\, \Phi_i}$$

4 – Crack control

Crack width w_k



Inputs
M_{Eqp} , b , h , d , d ', A_s , A'_s

Mean strain in rebars and concrete between 2 cracks

$$\varepsilon_{sm} - \varepsilon_{cm} = \frac{\sigma_s - k_t \dfrac{f_{ct,eff}}{\rho_{p,eff}} \left(1 + \alpha_e \, \rho_{p,eff}\right)}{E_s} \geq 0.6 \, \frac{\sigma_s}{E_s}$$

Maximum crack spacing

$$s_{r,max} = k_3 \, c + \frac{k_1 \, k_2 \, k_4 \, \Phi_{eq}}{\rho_{p,eff}}$$

Crack width

$$w_k = s_{r,max} \left(\varepsilon_{sm} - \varepsilon_{cm}\right)$$

Inputs
Exposure class

Limiting values of the crack width
w_{max}

$w_k \leq w_{max}$

NO → **Modified A_s**

YES → **OK**

5 – Deflection control

Flèches admissibles

Bon fontionnement des éléménts de la structure

Aspect inacceptable

Sous la combinaison quasi-permanente → $f \leq f_{max}$

f_{max} = flèche admissbale

f = flèche calculée de la poutre

Limiting deflection f_{max}

Criteria	Limiting deflection f_{max}
Aspect de fonctionnalité générale de la structure	l / 250
Endommage des éléments adjacents (cloisions, vitrages, reseaux…)	l / 500

l = span (m) → $l = l_{eff}$

5 – Deflection control

Limiting deflections

Pas d'obligation d'effectuer un calcul de flèche si l/d vérifie:

Assumptions: σ_s = 310 MPa for M_{Max} and $l_{eff} \leq 7$ m

Results from a parametric study made for beam and slab simply supported with rectangular cross section

$$\frac{l}{d} \leq K\left[11 + 1.5\sqrt{f_{ck}}\,\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{\frac{3}{2}}\right] \text{ if } \rho \leq \rho_0$$

$$\frac{l}{d} \leq K\left[11 + 1.5\sqrt{f_{ck}}\,\frac{\rho_0}{\rho - \rho'} + \frac{1}{12}\sqrt{f_{ck}}\sqrt{\frac{\rho'}{\rho_0}}\right] \text{ if } \rho \geq \rho_0$$

l = span (m) → $l = l_{eff}$

d = effective depth (m)

$\rho_0 = \sqrt{f_{ck}}\,10^{-3}$ → Reference reinforcement ratio (f_{ck} in MPa)

$\rho = A_s/(b\,d)$ → Tension reinforcement ratio due to M_{Max}

$\rho' = A_s'/(b\,d)$ → Compression reinforcement ratio due to M_{Max}

K → Factor to take into account the different structural systems

5 – Deflection control

Limiting deflections

Basic ratio of span (l) / effective depth (d) → K

Système structural	K	l/d	
		béton fortement sollicité $\rho \geq 1{,}5\ \%$	béton faiblement sollicité $\rho \leq 0{,}5\ \%$
Poutre sur appui simple	1,0	14	20
Dalle sur appui simple portant dans une direction		25	30
Travée de rive d'une poutre continue	1,3	18	26
Travée de rive d'une dalle continue portant dans une direction ou continue le long d'un grand coté et portant dans deux directions		30	35
Travée intermédiaire d'une poutre	1,5	20	30
Travée intermédiaire d'une dalle portant dans une ou deux directions		35	40
Dalle sans nervure sur poteaux (plancher-dalle) - pour la portée la plus longue	1,2	17	24
Poutre en console	0,4	6	8
Dalle en console	0,4	10	12

NF EN 1992 (Avril 2003). Eurocode 2 : Calcul de structures en béton – Partie 1.1 : Règles générales et règles pour les bâtiments – Annexe nationale. AFNOR.

5 – Deflection control

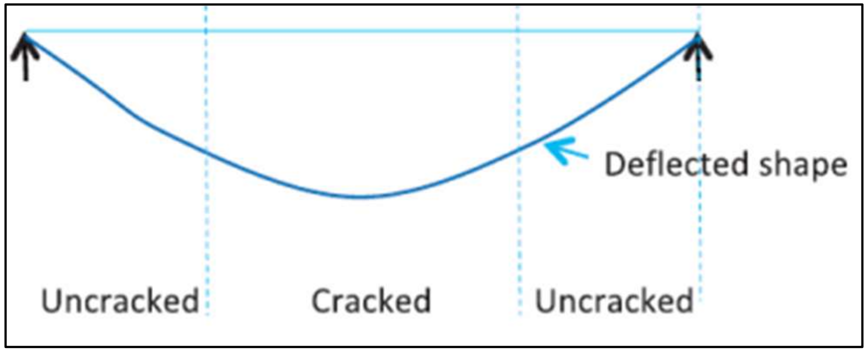
Limiting deflections

Deflection by calculation

Méthode rigoureuse EC2

Compute the deflection for the whole member twice
→ In fully cracked and uncracked conditions

1. **Discrétisation de la pouter en plusieurs tronçons**
2. **Calcul de la courbure dans chaque tronçon**
3. **Calcul de la flèche par integration numérique**



Simplified method in EC2

Calculate:

1. **The deflection in uncracked condition (tensile concrete take into account → I_h) : f_I**
2. **The deflection in fully cracked condition (tensile concrete neglected → I_{hr}) : f_{II}**
3. **The intermediate deflection between cracked and uncracked conditions : f**

I_h : Homogenous second moment of area (Inertie homogème)

I_{hr} : Reduced homogenous second moment of area (tensile concrete neglected)

5 – Deflection control

Deflection by calculation

$$f = \xi \, f_{II} + (1 - \xi) \, f_{I}$$

$$\xi = 1 - \beta \left(\frac{\sigma_{sr}}{\sigma_s}\right)^2 \qquad \text{or for flexure} \quad \xi = 1 - \beta \left(\frac{M_{cr}}{M_{Eqp}}\right)^2$$

= distribution coefficient
= 1 for a single short loading (courte durée)
= 0.5 for sustained load or many cycles of repeated loading (common case
= stress in tension reinforcement (cracked section)
= stress in tension reinforcement causing the 1st crack (cracked section)
= moment at SLS for quasi-permanent combination
= cracking moment for a maximum concrete tension stress σ_{ct} = $f_{ctm,fl}$

ξ = distribution coefficient

β = 1 for a single short loading (courte durée)

β = 0.5 for chargement prolongé or many cycles of repeated loading (common case)

σ_s = stress in tension reinforcement (cracked section)

σ_{sr} = stress in tension reinforcement causing the 1st crack (cracked section)

M_{Eqp} = moment at SLS for quasi-permanent combination

M_{cr} = cracking moment for a maximum concrete tension stress σ_{ct} = $f_{ctm,fl}$

$$f_{ctm,fl} = \mathrm{Max}\left\{\left(1.6 - \frac{h}{1000}\right) f_{ctm} \, ; \, f_{ctm}\right\} = \frac{M_{cr}\,(d - x)}{I_h} \rightarrow M_{cr} = \frac{f_{ctm,fl}\, I_h}{d - x}$$

h = overall depth of the cross-section in mm

x = depth of the concrete compression zone (uncracked section)

5 – Deflection control

Limiting deflections

Deflection by calculation

Reminder of beam theory for calculate the maximum deflection

Poutre sur 2 appuis simples – Charge uniformément répartie p

$$f_{max} = \frac{5\ p\ L^4}{384\ E\ I}$$

Poutre en console – Charge uniformément répartie p

$$f_{max} = \frac{p\ L^4}{8\ E\ I}$$

Where L = span

E = elastic modulus

I = second moment of area